



SIMATS Online Programming Lab

C PROGRAMMING



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CO5 - AT1 - LEVEL-3

📖 Level Exam

</> C

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Result Available

Your score: **70 / 100**

QUESTIONS

LEVEL 3 — HARD

Q1

function

Q2

functionsandpointers

2 / 2 saved

Level 3 — Hard **magic number (H)** — function

Write a function `int isMagicNumber(int n)` that checks if a given number `n` is a magic number (a number which repeatedly sums its digits until a single digit is obtained and that digit is 1). The function should return 1 if the number is a magic number, and 0 otherwise.

Sample Input:

Enter a number: 19

Sample Output:

19 is a magic number

Your Code

```
1 #include <stdio.h>
2 int isMagicNumber(int n)
3 {
4     while (n > 9)
5     {
6         int sum = 0;
7         while (n > 0)
8         {
9             sum += n % 10;
10            n /= 10;
11        }
```

```
12         n = sum;
13     }
14     return (n == 1) ? 1 : 0;
15 }
16 int main()
17 {
18     int n;
19     scanf("%d", &n);
20     int original = n;
21     if (isMagicNumber(n))
22         printf("%d is a magic number\n", original);
23     else
24         printf("%d is not a magic number\n", original);
25     return 0;
26 }
```

Program Input (stdin)

Enter input if required...

Output

19 is a magic number

 Run Save

Next →